

3. Project Planning Phase

3.1 Introduction

The project planning phase establishes the objectives, scope, resources, activities, schedule, and risks of ComicCraft. Its purpose is to create a clear roadmap so that development activities are organized, resources are used effectively, and the project is completed within the specified time.

3.2 Project Objectives

- To develop an easy-to-use AI-powered comic story creator.
- To provide secure user registration and authentication.
- To integrate Google Gemini models for script writing and panel image generation.
- To maintain consistent characters and art style across panels.
- To allow editing of scripts, panels, and speech bubbles.
- To export finished comics as PDF or images.
- To store saved comics in a personal gallery.

3.3 Project Scope

The scope includes the design and development of a web-based comic creation assistant. Users can create an account, log in, enter a story idea, generate a panel script, manage characters, generate panel artwork, edit speech bubbles, arrange pages, export the comic, and view saved comics. Animation, video generation, real-time collaboration, print-on-demand, and mobile apps are outside the current scope and are listed as future enhancements.

3.4 Project Modules

- Authentication Module
- Story Input Module
- AI Script Generator Module
- Character Management Module
- Panel Image Generation Module
- Speech Bubble and Caption Editor Module
- Layout and Export Module
- Gallery and History Module

3.5 Resource Planning

Resources include the development computer, internet connection, Python and FastAPI, a web frontend, a relational database, file storage for images, a Google Gemini API key with access to text and image models, Pillow/ReportLab, Visual Studio Code, and Git/GitHub for version control.

3.6 Project Schedule

Phase	Activities	Duration
1	Brainstorming, problem identification, idea finalization	Week 1
2	Requirement analysis and documentation	Week 1-2
3	System design, architecture, database design	Week 2-3
4	Backend and frontend development, Gemini text and image integration	Week 3-6
5	Testing and bug fixing	Week 6-7
6	Demonstration preparation and final documentation	Week 7-8

3.7 Risk Analysis

Risk	Impact	Mitigation
Gemini API downtime or rate limits	Script or image generation temporarily unavailable	Error handling, retry logic, queueing, user-friendly messages
Inconsistent character appearance	Comic looks disjointed	Stored character descriptions reused in every prompt; reference sheet; regenerate option
Unsafe or blocked content	Generation fails or inappropriate output	Input validation, Gemini safety settings, clear messages, retry with edited prompt
Text rendered inside images is unreliable	Garbled lettering in panels	Generate artwork without text; overlay dialogue as editable bubbles
Copyright concerns with styles or characters	Legal or ethical risk	Avoid prompts naming copyrighted characters; guidance and disclaimer to users
API key exposure	Security and misuse risk	Store key in server environment variables
Slow image generation	Poor user experience	Progress indicators, per-panel generation, lower resolution preview
Schedule delays	Incomplete features	Prioritize core modules first